

Nazarbayev University
Alejandro Melo Ponce

Fall 2019

Problem Set for Tutorial 3

To be discussed on T3, September 20, 2019

1. The market for paper is perfectly competitive and 1000 firms produce paper. The table sets out the market demand schedule for paper.

Price (dollars per box)	Quantity (thousands of boxes per week)
3.65	500
5.20	450
6.80	400
8.40	350
10.00	300
11.60	250
13.20	200

The next table sets out the costs of each producer of paper.

Output (boxes per week)	Marginal cost (dollars per additional box)	Average variable cost (dollars per box)	Average total cost
200	6.40	7.80	12.80
250	7.00	7.00	11.00
300	7.65	7.10	10.43
350	8.40	7.20	10.06
400	10.00	7.50	10.00
450	12.40	8.00	10.22
500	20.70	9.00	11.00

- (a) Calculate the market price, the market output, the quantity produced by each firm, and the firm's economic profit or loss.
- (b) Now suppose that the market demand decreases and the demand schedule becomes:

Price (dollars per box)	Quantity (thousands of boxes per week)
2.95	500
4.13	450
5.30	400
6.48	350
7.65	300
8.83	250
10.00	200
11.18	150

If firms have the same costs as in the previous item, what is the market price and the firm's economic profit or loss in the short run?

2. The table provides data on a market demand schedule (top two rows) and a firm's average and marginal cost schedules (bottom four rows).

Price P	24	20	16	12	8
Quantity	3000	4000	5000	6000	7000
Firm's Output	1	2	3	4	5
MC	11	11.13	12.00	13.63	16.00
ATC	13.50	12.25	12.00	12.19	12.70
AVC	11.25	11.13	11.25	11.63	12.25

- What is the firm's shutdown point?
- If there are 1000 identical firms in the market, what is the market price and quantity?
- With 1000, will firms enter or exit?
- Calculate the long-run equilibrium price, quantity, and number of firms.